AN

INTRODUCTION

TO

NUMERICAL METHODS

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AUTHOR OF DATA STRUCTURES AND ALGORITHMS.
This book is the same as the last series is just a way of taking notes and to learn Numerical Methods

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Just me taking notes with LaTeX

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PREFACE

This paper continues my dedicated method of note-taking, though it diverges from previous works as it focuses on my current academic studies. This subject is of paramount importance to me, as it explores the mathematical expressions that underpin algorithms. It serves as a companion to my ongoing project, the "Bible of Algorithms," providing a rigorous look at the theoretical origins of these formulas and how they are translated into mathematical models. These notes are based on the lectures and seminars of Professor Fatmir Hoxha, a distinguished mathematician who earned his PhD in France. Coming from a tradition that deeply respects the beauty of mathematics, I find the intersection of math and computer science truly profound. A fascinating realization in this field is that a number is rarely "exact" in a machine; a value we call 2 is often represented as 2.0000000000134... or 1.999999999... in floating-point arithmetic. This book explores that very challenge: the art of approaching the absolute truth by minimizing numerical error as much as possible.

Mathematical Analysis	Numerical Analysis
1. Analytical Methods: These provide direct algorithms that lead to a unique, exact solution (a).	1. Numerical Methods: These utilize iterative algorithms to generate approximate solutions ($x^* \approx a$).
2. **Absolute Accuracy**: Operates with 100% precision where $\exists f(x)$ is evaluated symbolically.	2. **Error Tolerance (ϵ)**: Based on numerical data where we aim for the smallest possible error ($\epsilon = 0.0001$).
3. Discrete results are rare; focuses on continuous logic.	3. Iterative results; depends on numerical data sets and machine precision.

I. Fundamental Definitions

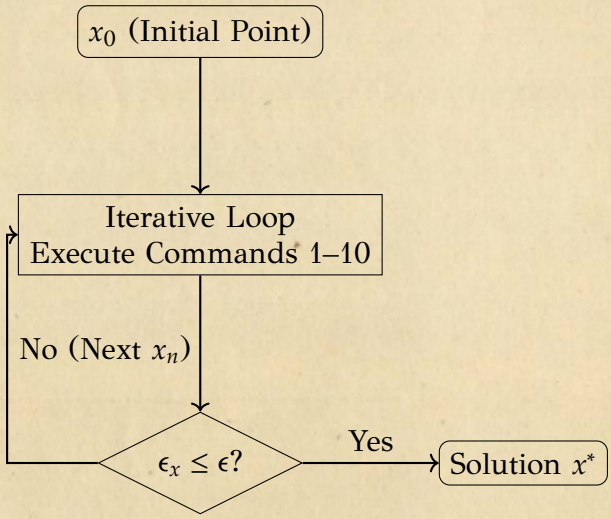
Numerical computation begins with understanding the nature of instructions and their inherent inaccuracies:

- **Algorithm (Algoritmi)**: A finite sequence of instructions designed to solve a problem (*Varg i fundëm instruksionesh*).
- **Numerical Error (Gabimi)**: The deviation between the true value and the value produced by the machine (*Ndryshimi midis madhësisë dhe asaj që njehsojmë*).
- **The Constant π**: In computation, 3.14 is a truncation. The "real" π contains infinite info that a machine cannot store.

II. The Iterative Process

Iterative methods generate a sequence $\{x_1, x_2, \ldots, x_n\}$ that approaches the limit a.

- **Initial Guess (x_0)**: The "përafrimi fillestar" or starting approximation.
- **Convergence**: A method is successful if each iteration is "better" than the previous: $|x_{n+1} - a| < |x_n - a|$.
- **Termination Criteria**: The process stops when $|x_n - x_{n-1}| \leq \epsilon$.



III. Theory of Errors

Errors are categorized based on their relation to the true value x and the approximation X (or x^*).

1. Absolute Error (Δ)

The simplest measure of inaccuracy is the magnitude of the difference:

$$\Delta = |x - X| \leq \Delta_x$$

Here, Δ_x is the *Absolute Error Bound*. This implies the true value x lies in the interval:

$$X - \Delta_x \leq x \leq X + \Delta_x \tag{1}$$

2. Relative Error (δ)

This scales the absolute error by the size of the approximation, usually expressed as a percentage:

$$\delta = \frac{\Delta}{|X|} \leq \delta_x \implies \Delta \leq |x|\delta_x$$

IV. Significant Digits and Precision

Significant Digits (Shifrat me vlerë): These are all digits starting from the first non-zero digit on the left. *Example:* In 0.00**70120500**, the significant digits are 7, 0, 1, 2, 0, 5, 0, 0.

The Positional Expansion Method:
To find the position m of the first significant digit, we decompose the number:

$$325.17 = 3 \cdot 10^2 + 2 \cdot 10^1 + 5 \cdot 10^0 + 1 \cdot 10^{-1} + 7 \cdot 10^{-2}$$

In this case, $m = 2$.

Theorem of Correct Digits (n):
A number has n correct digits if the absolute error Δx satisfies:

$$\Delta x \leq \frac{1}{2} \cdot 10^{m-n+1}$$

Practical Exercise (from Notebook):
Given $x = 325.17351$ with $\Delta_x = 0.004$. Determine the number of correct digits n.

1. We know $m = 2$ from the expansion above.
2. We define a safe bound: $\Delta_x < 0.005 = \frac{1}{2}10^{-2}$.
3. Solve for n: $m - n + 1 = -2 \implies 2 - n + 1 = -2 \implies n = 5$.
4. **Result**: The first 5 digits $(3, 2, 5, 1, 7)$ are correct.

Rounding Strategy (Rruga e Shkurtër):
When $a = 12.076$ and $\Delta_a = 0.0005 = \frac{1}{2}10^{-3}$: The error occurs at the 10^{-3} position. Digits at higher positions are correct.

- Significant digits: $1, 2, 0, 7, 6$.
- If we round to 3 significant digits: $176.0324 \to 176.03$.

V. Error Propagation in Functions

When a result z is derived from multiple variables $z = f(x_1, x_2, \ldots, x_n)$, the error propagates through partial derivatives:

$$\Delta z \approx \left|\frac{z}{x_1}\right| \Delta x_1 + \left|\frac{z}{x_2}\right| \Delta x_2 + \cdots + \left|\frac{z}{x_n}\right| \Delta x_n \tag{2}$$

Example Function from Lecture:

$$z = e^{\sqrt{x_1^2 + \sin(x_2)}} - 3$$

The absolute error Δz is the sum of individual sensitivities Δx_i.

VI. The Cancellation Error (Gabimi i Fshirjes)

This error occurs when subtracting two nearly equal numbers ($x_1 \approx x_2$). *Example:* Let $x_1 = 50.137$ and $x_2 = 50.127$ with $\Delta x = 0.0005$.

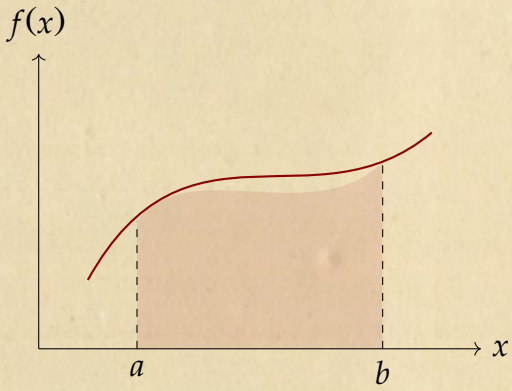
- Difference $u = |x_1 - x_2| = 0.010$.
- Absolute Error $\Delta u = \Delta x_1 + \Delta x_2 = 0.001$.
- Relative Error $\delta_u = \frac{0.001}{0.010} = 0.1 = 10\%$.

Despite the initial values being very accurate, the result loses almost all significant digits. **Solution**: Use Double Precision (*Saktësi dyfishe*).

VII. Introduction to Numerical Integration

When an integral $\int f(x)dx$ has no analytical solution, we use weighted sums:

$$\int_a^b f(x)dx \approx \sum_{i=0}^{n} A_i f(x_i)$$



VIII. Final Practice Set

Round the following numbers to 3 significant digits and calculate Δx and δx:

1. $x_1 = 2.1514 \to X_1 = 2.15$.
2. $x_2 = 0.16152 \to X_2 = 0.162$.
3. $x_3 = 0.01204 \to X_3 = 0.0120$.

End of Chapters: Error Theory and Iterative Foundations.